GeauxTigers_PLAN.md

Project 3: Group Research Plan
KIN 7518 Social Issues in Sport | Due: April 10, 2026

1. Research Questions & Significance

RQ 1

The Question: Do athletes face more positive or negative engagement when making politically charged posts on social media?

The Context: Athletes today aren't just known for what they do on the field, court, etc.—they also have a huge presence on social media. Platforms like Instagram, Twitter, and YouTube give them the ability to share their opinions directly with fans, including on political and social issues. When athletes post politically charged content, it usually gets a lot of attention, especially in the comments. The audiences' opposing views can turn these posts into spaces where people react and can spark debate or controversy and in some cases call into question an athlete's character.

Why It Matters: Social media is one of those main places where people form and share their opinions, so it plays a big role in how athletes are viewed outside of their sport. Looking at whether politically charged posts get more positive or negative engagement helps us understand how much fans let politically motivated posts influence their view on the athlete professionally. It also shows whether people expect athletes to stay focused on their sport or to use their platform to speak out about bigger issues.

2. Dataset Selection & Justification

Dataset Choice: Politics in Sport

Justification: Instagram comment data is a strong fit for this research because it provides direct insight into how audiences react to athletes' politically charged posts in real time. Since Instagram is one of the main platforms athletes use to share personal opinions and engage with fans, it serves as a relevant space to study public response.

Key file used: B50_INS_COMMENT (Sheet1)

3. Preliminary Variable Operationalization

The lists were built using three combined methods: (1) bottom-up word frequency analysis of the 1,717 politically engaged comments to identify what words fans actually used; (2) the AFINN sentiment lexicon, a public-domain word-scoring system developed for social media text, which provided validated positive/negative word coverage far beyond what manual brainstorming can produce; and (3) iterative expansion, where the most frequent unscored words in each tone group were reviewed and added if they carried clear sentiment value. Political keywords terms like "kamala," "vote," "democrat," and "republican" that appeared in the data were also included.

Construct	Operational Definition	Data Source / Indicator

Politically engaged comment	Comment containing at least one predefined political keyword. Binary coded: 1 = present, 0 = absent.	text field; case-insensitive keyword search applied to full comment string
Positive tone	Politically engaged comment with at least one positive keyword and no negative keyword.	Keyword search on text field within the politically engaged subset
Negative tone	Politically engaged comment with at least one negative keyword and no positive keyword.	Keyword search on text field within the politically engaged subset
Mixed tone	Politically engaged comment containing both positive and negative keywords simultaneously.	Keyword co-occurrence within a single comment
Neutral tone	Politically engaged comment with neither positive nor negative keywords — political in content but sentiment-unclassified.	Residual category after positive/negative/mixed assignment
Engagement	Raw like count per comment and comment presence in dataset.	likes field in B50_INS_COMMENT

Keyword lists:

Category	Keywords	Count
Political (18 terms)	"trump," "president," "biden," "maga," "obama," "reagan," "political," "supporter," "american," "kamala," "joebiden," "election," "vote," "republican," "democrat," "conservative," "liberal," "january 6"	1,848 comments
Positive (47 terms)	"love," "loved," "best," "awesome," "cool," "amazing," "great," "goat," "legends," "iconic," "terrific," "fantastic," "excellent," "wonderful," "superb," "outstanding," "brilliant," "incredible," "phenomenal," "spectacular," "magnificent," "glorious," "legendary," "perfect," "good," "nice," "fun," "enjoy," "enjoyed," "happy," "glad," "pleased," "proud," "respect," "impressive," "beautiful," "smart," "brave," "excited," "exciting," "inspired," "inspiring," "talented," "winner," "winning," "win," "champion," "hero," "wow," "favorite," "lfg," "congrats," "congratulations," "well done," "blessed," "bravo"	414 positive comments
Negative (49 terms)	"lose," "vomit," "gross," "suck," "unfollow," "dump," "ew," "hate," "disgusting," "horrible," "terrible," "awful," "pathetic," "shameful," "trash," "stupid," "idiot," "moron," "loser," "losers," "corrupt," "evil," "disgrace," "appalling," "worst," "bad," "poor," "wrong," "fail," "failed," "failure," "liar," "lie," "lies," "fake," "embarrassing," "sad," "disappointing," "disappointed," "ridiculous," "absurd," "nonsense," "disrespect," "disrespectful," "unfair," "dishonest," "boring," "cheat," "cheats," "cheating," "clown," "joke," "fraud," "criminal," "cringe"	183 negative comments

Observed counts:

Tone	Count	%	Avg Likes	Eng.-Weighted
Neutral	1,162	62.9%	20.82	24,192
Positive	414	22.4%	6.97	2,886
Negative	183	9.9%	4.37	800
Mixed	89	4.8%	3.91	348
Total political	**1,848**	**100%**	—	—

4. Proposed Analyses

Keyword frequency analysis

We tallied the raw occurrence count of each political, positive, and negative keyword across all 11,833 comments. Among political keywords, "trump" was by far the most frequent (1,121 occurrences), followed by "biden" (326) and "president" (324). Among positive keywords in the political subset, "good" (102), "great" (80), and "love" (78) led; among negative keywords, "ew" (67), "lie" (61), and "hate" (37) appeared most often. This analysis directly addresses RQ1 by showing which political figures and sentiment terms dominated the comment space.

Cross-tabulation: tone x engagement

A cross-tabulation of tone category against engagement metrics (like count, comment presence) allows comparison of how each sentiment type performs. With these keywords, neutral political comments receive the highest average likes (20.82) compared to positive (6.97) or negative (4.37), suggesting that measured or observational reactions attract broader endorsement than overtly charged ones. This directly tests whether politically charged content generates measurably positive or negative audience response.

Engagement-weighted frequency

Each tone category was weighted by multiplying its comment count by average likes received (count x avg_likes). Using these counts: neutral scored 24,192, positive 2,886, negative 800, and mixed 348. This approach addresses RQ1 by showing not just how many comments fall into each tone, but how much those comments resonated with broader audiences beyond the original commenter.

Temporal trend analysis

Comments were grouped by month of posting. Of the 1,848 politically engaged comments, 1,679+ concentrated in July 2024, indicating a strong event-driven spike likely aligned with a high-profile athlete post or political moment. Tracking political engagement over time helps determine whether spikes are event-driven rather than a stable baseline, which has implications for generalizing findings.

5. Limitations & Potential Issues

Platform bias: Instagram users are typically younger and more active online, so the reactions we see might not represent all sports fans.

Engagement does not equal sentiment: Likes and comments don't always clearly show if something is positive or negative. A post with many comments could be controversial rather than supported.

Sentiment challenges: It can be hard to tell tone in comments (sarcasm, jokes) so some responses could be misclassified. Even with an abundance of keywords, neutral at 63% may partly reflect irony or ambiguity that keyword matching cannot resolve.

Silent majority: Most people don't comment or like posts, so this reflects only more vocal users who may have stronger opinions.

Context matters: Reactions can depend on the topic, timing, or world events. The concentration of political comments in July 2024 suggests event-driven clustering that may not generalize to other time periods.

Researcher perspective: What we label as positive or negative is somewhat subjective. Coding decisions are clearly defined, applied consistently via automated search, and spot-checked manually.

Remaining keyword gaps: The lists remain a simplified proxy for sentiment. Terms unique to specific political events, platform-specific slang, and non-English comments are not captured. Future work could apply a pre-trained classifier (e.g., VADER or a fine-tuned BERT model) to the full comment corpus for more comprehensive coverage.

6. Ethical Considerations

Privacy: All data comes from public Instagram comments. No private information is used, and usernames are not included in any reported outputs to preserve anonymity. The dataset contains user IDs and handles, but these are not referenced in analysis or reporting.

Harm: This project focuses on aggregate trends, not individual people. We avoid making broad assumptions about fan groups or political affiliations and are mindful of how findings are framed. Because the dataset contains comments referencing real political figures, we are careful not to misrepresent public opinion or draw causal conclusions beyond what the data supports.

Bias: There is built-in bias from Instagram's algorithm and from how engagement is operationalized through likes and comment presence. Our keyword-based sentiment coding is a simplified proxy for actual sentiment. To reduce interpretive bias, coding decisions are explicitly defined in Section 3, applied consistently via automated search, and spot-checked manually against random comment samples.

7. Group Role Assignments

Member	Role	Responsibilities

Antoine	Data Lead	Data cleaning, keyword list construction (including bottom-up frequency analysis and AFINN expansion), frequency counts, file management

Areli	Theory Lead	Literature integration, RQ refinement, framing interpretation, written synthesis
Ethan	Methods Lead	Analysis design, coding scheme development, tool implementation, cross-tabulation

8. Data Visualization Plan

Primary Goal: The visualization answers the core question of RQ1: do politically engaged comments skew positive or negative? It also shows which keywords drove the most volume and how average likes differed across sentiment categories.

Visualization Description: Three coordinated charts: (1) a donut chart showing the tone distribution (positive/negative/neutral/mixed) among the 1,848 politically engaged comments; (2) a bar chart comparing average likes per comment across the four tone categories; and (3) a horizontal bar chart of political keyword frequency across all 11,833 comments.

Design Rationale: The donut chart efficiently communicates proportional breakdown at a glance. The average-likes bar chart pairs with it to show that comment volume and audience validation do not move in the same direction — a key finding. The keyword frequency chart grounds the analysis in the actual vocabulary of fans. Together, the three panels address both the volume and quality of engagement, which are distinct but related dimensions of RQ1.

Chart 1 · Tone Distribution Among Politically Engaged Comments

Share of the 1,848 politically engaged comments by assigned sentiment category.

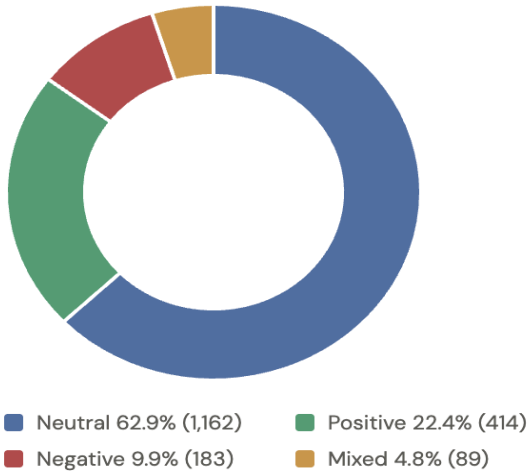


Chart 2 · Average Likes per Comment by Tone Category

Mean like count for each tone group. Higher likes suggest broader audience endorsement beyond the original commenter.

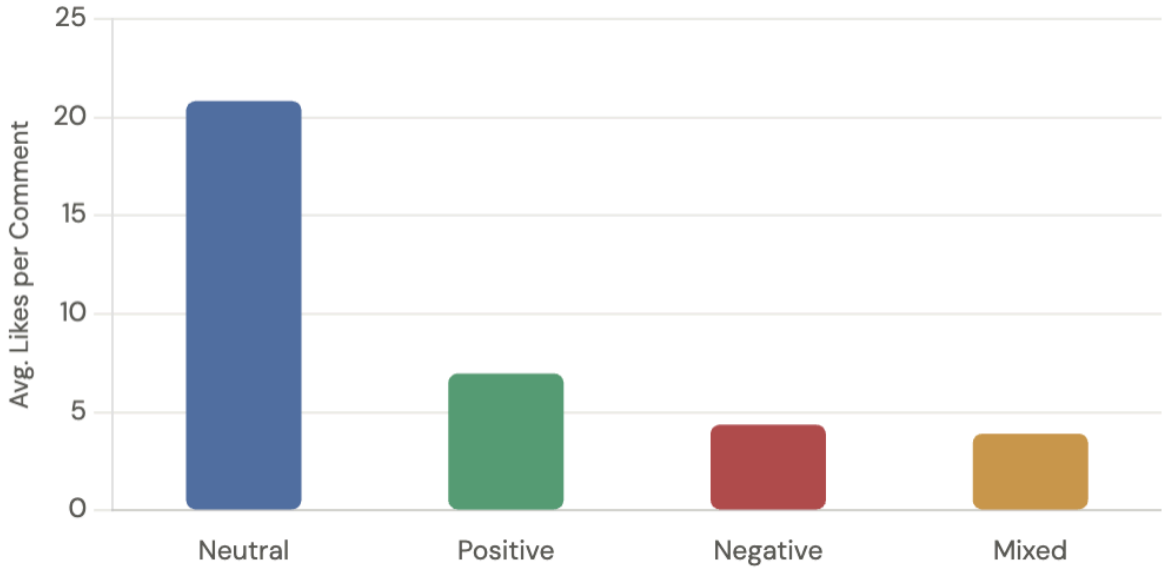
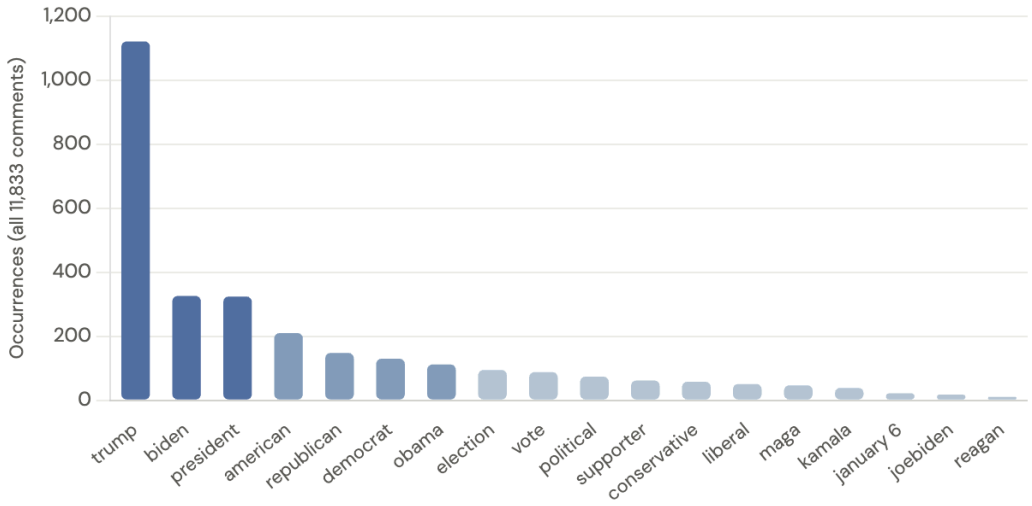


Chart 3 · Political Keyword Frequency Across All 11,833 Comments

Raw occurrence count of each of the 18 political keywords. A single comment may contain multiple keywords.



Verification Methods:

• Spot-checked keyword counts against manual review of random comment samples • Had groupmate review keyword lists for face validity before running analysis • Verified tone category totals sum to 1,848

• Cross-checked engagement-weighted scores: count x avg_likes = reported weighted value (e.g., neutral: 1,162 x 20.82 = 24,192)

• Confirmed total comment count (11,833) matches raw CSV row count

Summary data table — keyword lists:

Tone	Count	% of Political	Avg Likes	Eng.-Weighted Score
Neutral	1,162	62.9%	20.82	24,192
Positive	414	22.4%	6.97	2,886
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Mixed	89	4.8%	3.91	348
Total political	**1,848**	**100%**	—	—

Brief Interpretation: Using these keyword lists, politically engaged comments constitute 15.6% of all 11,833 comments (1,848 total). Among those, 62.9% are neutral in tone, 22.4% are positive, 9.9% are negative, and 4.8% are mixed. Positive comments outnumber negative by more than 2-to-1, suggesting

that when fans engage politically with athlete content they lean favorable. Neutral comments still attract the highest average likes (20.82 vs. 6.97 for positive and 4.37 for negative), indicating that observational or factual-sounding political comments — rather than emotionally charged ones — earn broader audience endorsement.

9. AI-Assisted Work Documentation & Verification

Tools Used: Claude (Anthropic, claude-sonnet-4-6) was used to assist with data analysis (Python/pandas keyword coding, bottom-up word frequency analysis, and AFINN-based lexicon expansion on B50_INS_COMMENT), generating data visualizations (Chart.js), and formatting the final document.

Verification Methods: All keyword counts and tone classifications produced by AI-assisted code were verified by manually reviewing a random sample of politically engaged comments and confirming their assigned tone categories matched the keyword coding logic. The engagement-weighted score for neutral was recalculated by hand (1,162 x 20.82 = 24,192) to confirm the formula was applied correctly.

Code Explanation: The Python analysis used pandas to load the CSV with encoding='latin1' (required due to non-UTF-8 characters in comment text). Each comment's text field was lowercased and searched for membership in the predefined keyword lists using Python's in operator. Political engagement was coded as a binary flag; tone was assigned by checking positive and negative keyword presence within the politically engaged subset using a four-way classification (positive-only, negative-only, both/mixed, neither/neutral). Keyword lists were analyzed using three combined methods: (1) word frequency analysis via Python's Counter on the politically engaged subset, (2) AFINN sentiment lexicon cross-referencing, and (3) iterative review of the most frequent unscored words in each tone group. Average likes were computed with .groupby('tone')['likes'].mean() and engagement-weighted scores as len(group) x mean_likes per tone category.

Learning Reflection: Working with real Instagram comment data reinforced how much the choice of keyword list shapes results. In past projects brainstormed lists can create figures that look like substantive findings but are largely an artifact of undercoverage. Combining bottom-up frequency analysis with an established lexicon (AFINN) allows for a more accurate detection of both positive and negative comments, which changes the story the data tells. This taught us that operationalization decisions are not just methodological formalities — they directly determine what you can and cannot see in the data, and they need to be grounded in evidence rather than assumption.